

Tafila Technical University

Faculty of Information Technology

Digital Forensics

Disk Image Analysis using Autopsy

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Abstract

This project presents a practical digital forensics investigation focused on **disk image analysis using Autopsy**. The objective of the project is to demonstrate how digital evidence can be identified and analyzed using standard forensic tools and methodologies. A controlled test dataset was prepared to simulate normal user activity, including file creation and deletion prior to analysis.

The investigation was conducted in a virtual environment to ensure forensic soundness and prevent alteration of the original data. A forensic image was created using FTK Imager, and the image integrity was verified through hash values. The disk image was then analyzed using Autopsy to classify file types and examine file metadata.

The results demonstrate the effectiveness of Autopsy in organizing digital evidence and providing useful insights into stored files and user activity. This project enhanced practical understanding of forensic workflows, disk image analysis, and the limitations of logical-level forensic examination in an educational context.

Introduction

Digital forensics is a specialized field within cybersecurity that focuses on the identification, preservation, analysis, and presentation of digital evidence. It plays a critical role in investigating cyber incidents, data breaches, and unauthorized activities by examining data stored on digital devices. One of the most important techniques in digital forensics is disk image analysis, which allows investigators to analyze data without modifying the original evidence.

This project focuses on disk image analysis using Autopsy, an open-source digital forensics tool widely used for educational and investigative purposes. Autopsy provides features for file classification, metadata analysis, and evidence organization. By performing this analysis in a controlled virtual environment, the project demonstrates practical forensic workflows while maintaining forensic integrity and adhering to standard digital forensics methodologies.

Objectives

The objectives of this project are:

- To create a controlled test dataset that simulates normal user file activity.
- To simulate user behavior by deleting a file prior to forensic analysis.
- To perform disk image analysis using the Autopsy digital forensics tool.
- To identify and classify files discovered during the forensic investigation.
- To examine file metadata and analyze its relevance to digital forensic investigations.

Methodology

○ Test Data Preparation

A controlled test dataset was prepared to simulate normal user file activity. A folder named *Forensics_Test* was created containing multiple file types, including text and document files. This step was performed to create a realistic environment for forensic analysis.

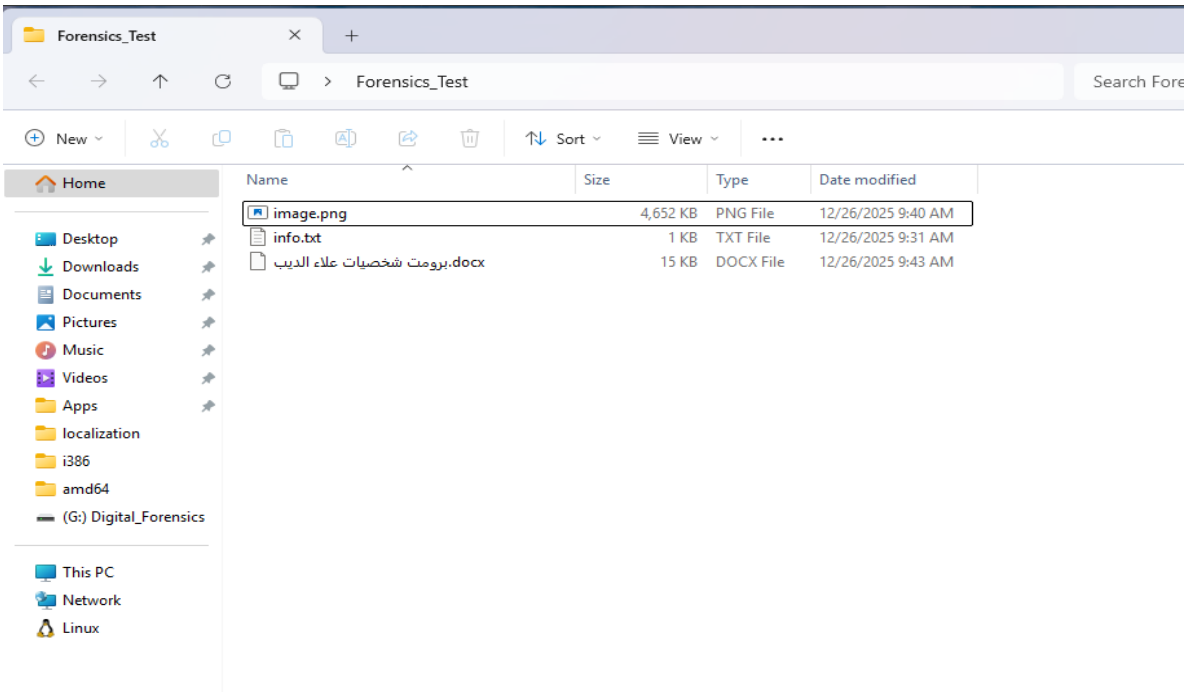


Figure 1: Test folder containing sample user files prepared for forensic analysis

To simulate user behavior and potential evidence manipulation, one file was deleted from the test folder prior to the forensic imaging process. This action represents a common scenario investigated in digital forensic cases.

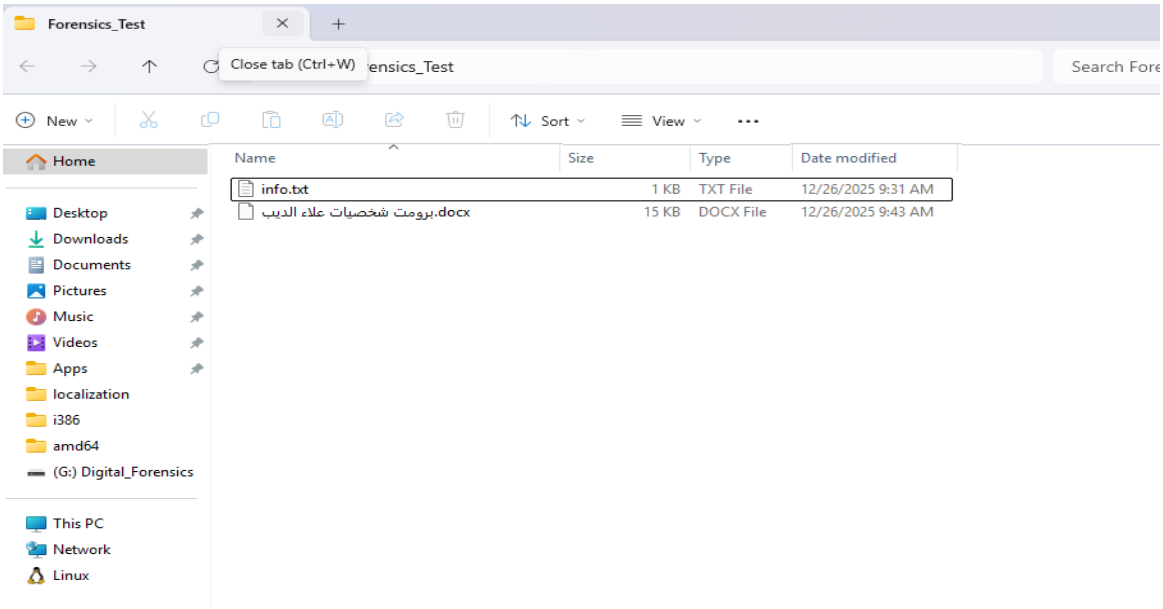


Figure 2: A file was deleted to simulate user activity prior to forensic imaging

- **Disk Image Creation Using FTK Imager**

- **After preparing the test data, a forensic image was created using FTK Imager. The imaging process was performed to preserve the data in its current state and prevent any modification during analysis. Logical imaging was used to capture the contents of the test folder.**

- **During this step, the source folder and destination location were selected, and image verification was enabled to ensure the integrity of the forensic image after creation.**

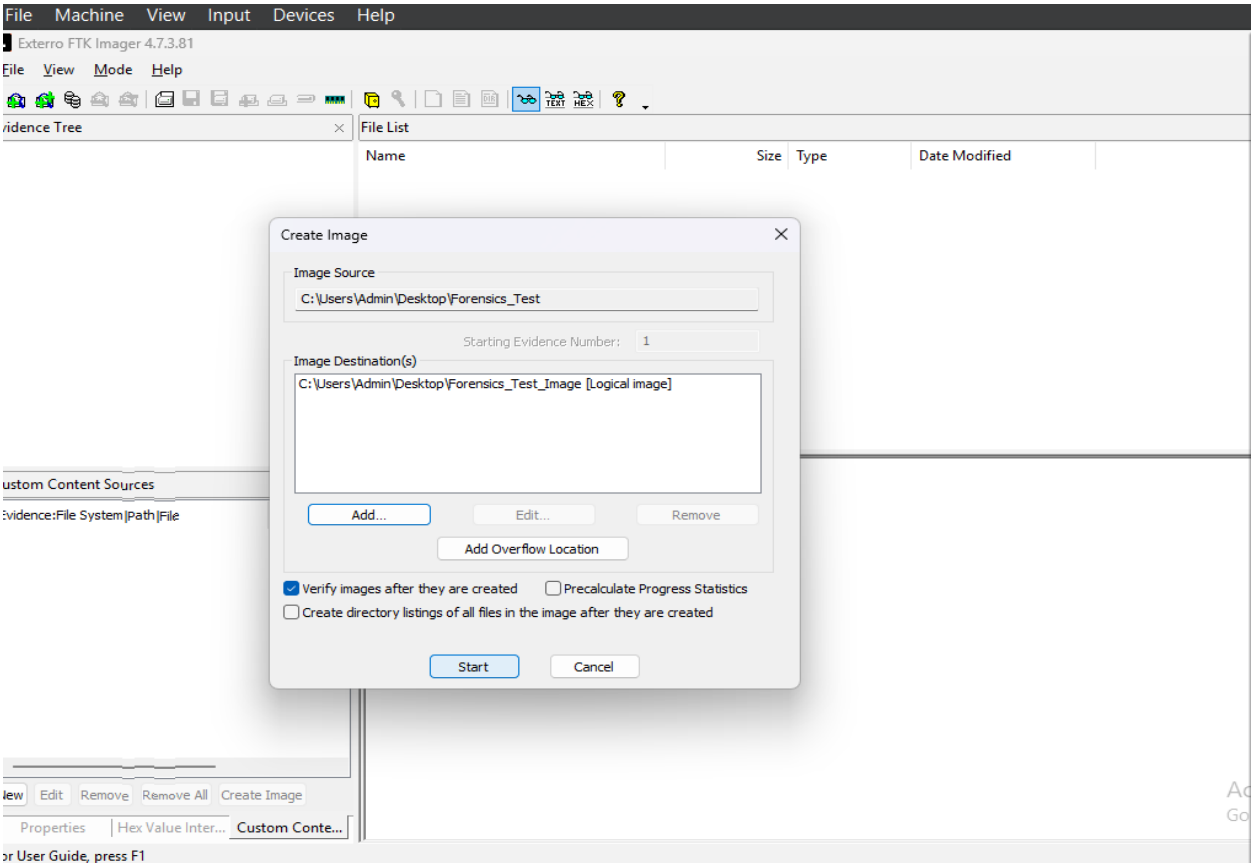


Figure 3: Selecting the source and destination during the creation of a forensic image using FTK Imager

- After the imaging process was completed, the forensic image was verified using cryptographic hash functions. The matching MD5 and SHA1 hash values confirm that the image integrity was preserved and that no data was altered during the acquisition process.

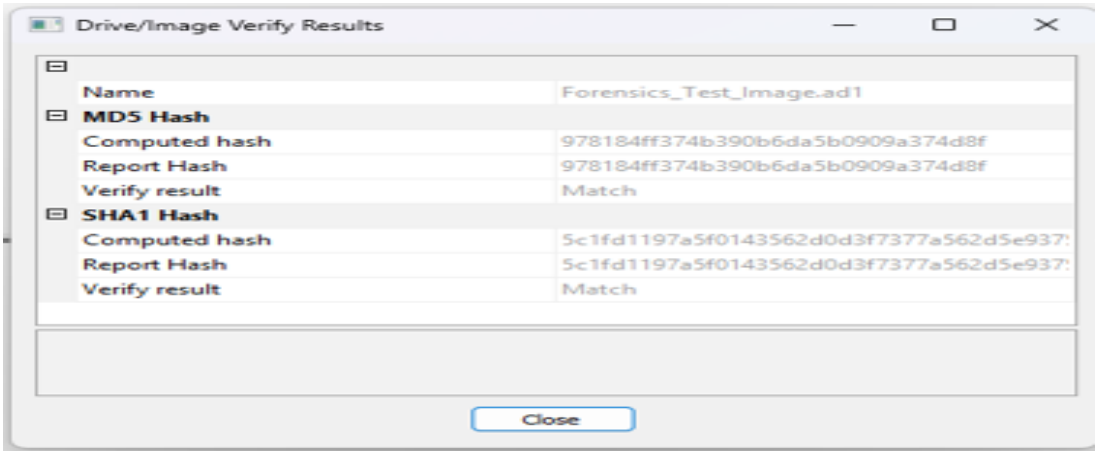


Figure 4: Verification results showing matching MD5 and SHA1 hash values for the created forensic image

- The imaging process was completed successfully using FTK Imager. The status message confirms that the forensic image was created without errors and is ready for further forensic analysis.

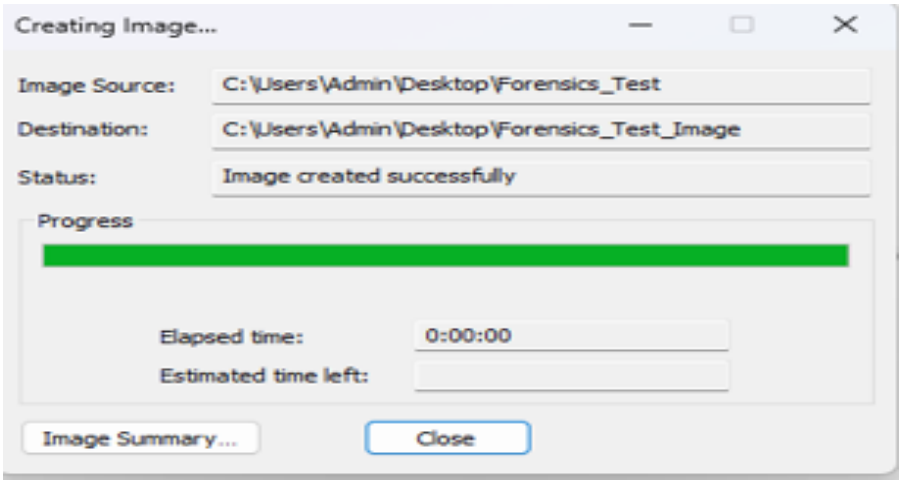


Figure 5: Successful creation of the forensic image using FTK Imager

Forensic Analysis and Findings

After completing the forensic image acquisition, the analysis phase was initiated using Autopsy. The forensic image was successfully added as a data source, allowing the examination of the stored files and artifacts.

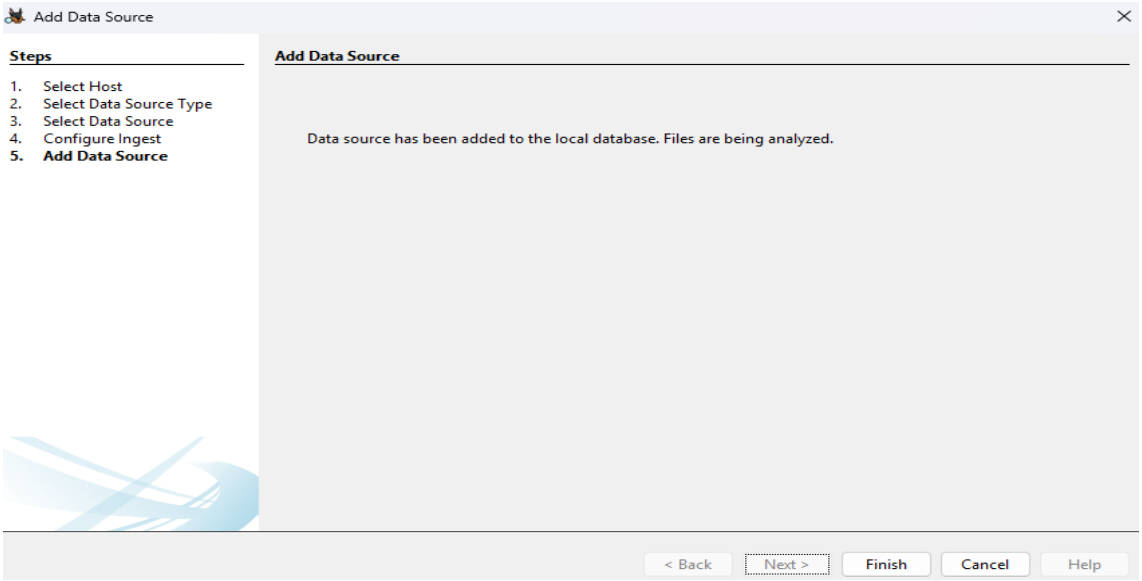


Figure 6: Successful addition of the forensic image as a data source in Autopsy and initiation of the analysis process

File Type Analysis

Autopsy organized the files within the forensic image based on their file types. This categorization helps in understanding the nature of the data stored in the image and assists investigators in focusing on relevant files during the analysis process.

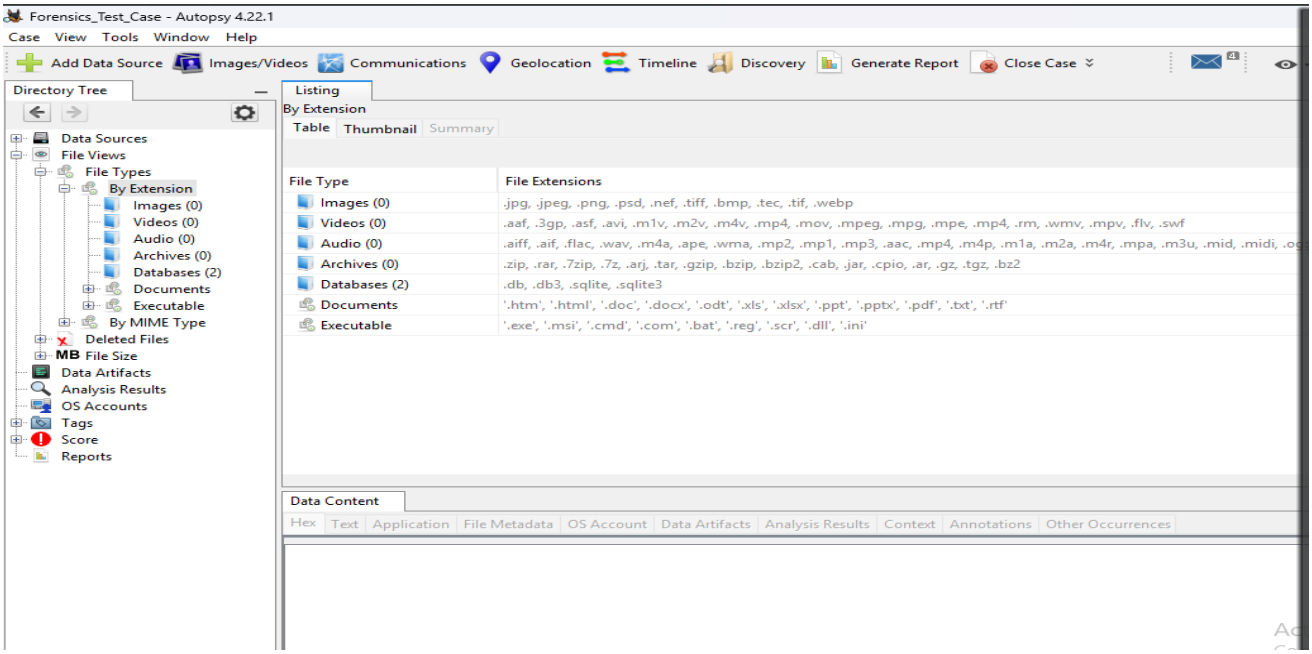


Figure 7: : File type classification results based on file extensions in Autopsy

File Metadata Analysis

- Autopsy was used to examine the metadata of the recovered text file (`info.txt`). Metadata analysis provides critical contextual information about a file, including its size, allocation status, timestamps, and file type. This information helps investigators understand how the file exists within the file system and assess its relevance to the investigation.
- The analysis showed that the file was allocated and stored as a plain text file. However, timestamp values (creation, modification, access, and change times) were unavailable or set to zero, indicating limitations in timestamp preservation due to the logical imaging process. Despite this, the file was successfully identified and located within the forensic image, confirming its presence and integrity during analysis.

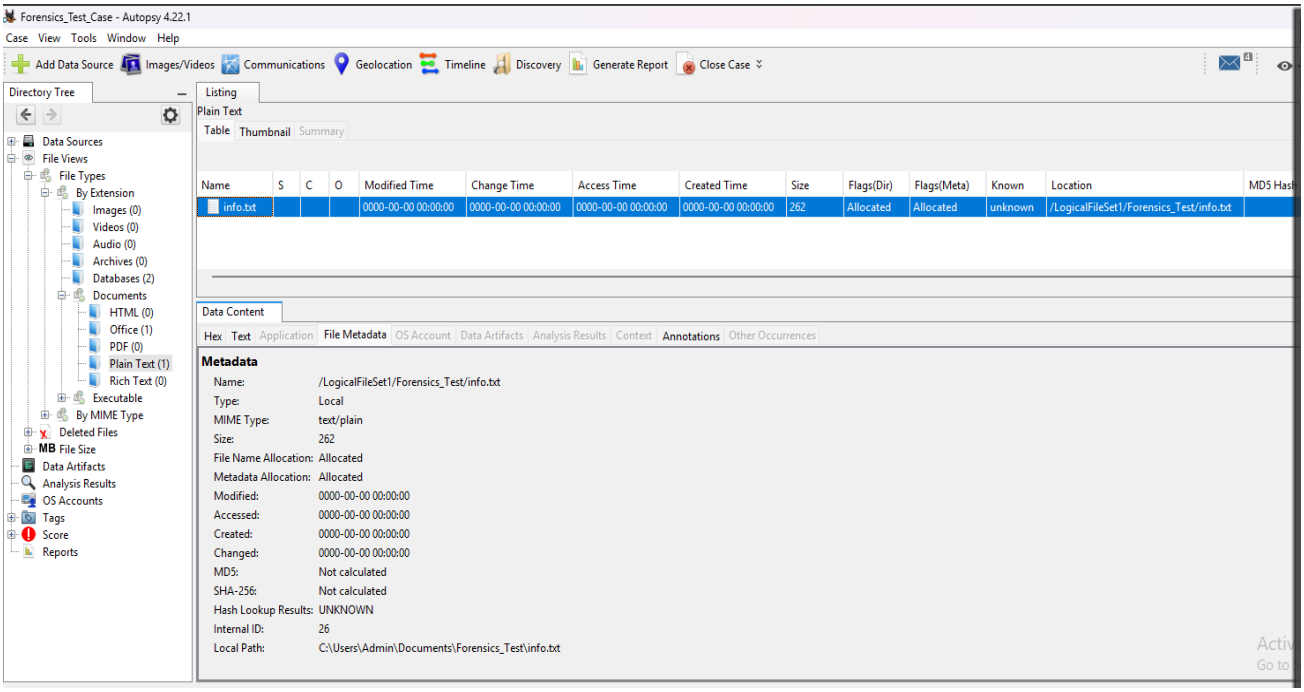


Figure 8: metadata of the analyzed text file (info.txt) using Autopsy

- This demonstrates how digital forensic tools can still extract valuable information from files even when certain metadata attributes are incomplete or missing.

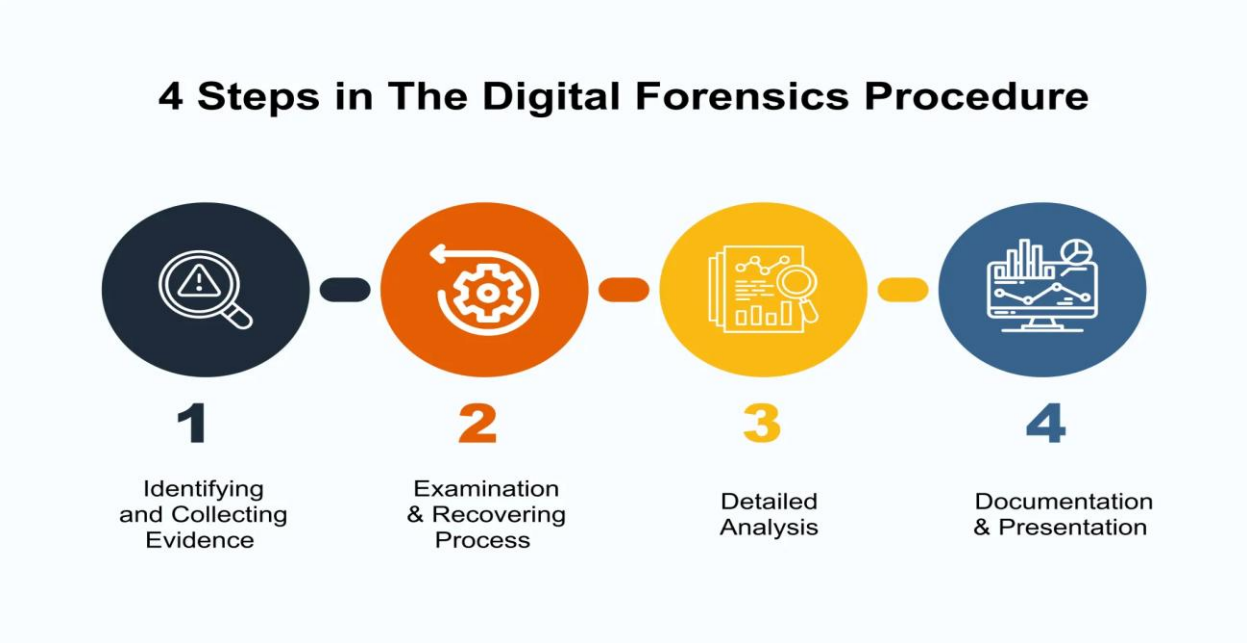


Conclusion

- This project demonstrated the practical application of digital forensic techniques for disk image analysis using Autopsy and FTK Imager. A logical forensic image was successfully created from the test dataset while ensuring data integrity through cryptographic hash verification (MD5 and SHA1). The matching hash values confirmed that the imaging process was performed correctly without altering the original data.

- Through forensic analysis in Autopsy, the contents of the image were systematically examined. Files were categorized by type, and metadata analysis was conducted to extract contextual information such as file size, allocation status, and file location. Although some timestamp values were unavailable due to limitations associated with logical imaging, the analysis successfully identified and examined the target file within the forensic image.

- Overall, this investigation highlights the importance of proper evidence acquisition, verification, and structured analysis in digital forensics. The use of industry-standard tools enabled effective examination of digital evidence and demonstrated how forensic methodologies can uncover and validate information even when certain metadata attributes are incomplete.



References:

No.	Reference	Description
1	Sleuth Kit & Autopsy. Autopsy User Documentation. https://www.autopsy.com	Official documentation used for digital forensic analysis and file metadata examination.
2	Exterro. FTK Imager User Guide. https://www.exterro.com/ftk-imager	Used for creating and verifying forensic disk images using logical imaging.
3	Casey, E. (2011). Digital Evidence and Computer Crime. Academic Press.	Reference for digital forensic principles and investigation methodology.
4	NIST. Guide to Integrating Forensic Techniques into Incident Response. NIST SP 800-86.	Provides standard forensic investigation practices and procedures.